

Mobile Architectures

Mobile Systems

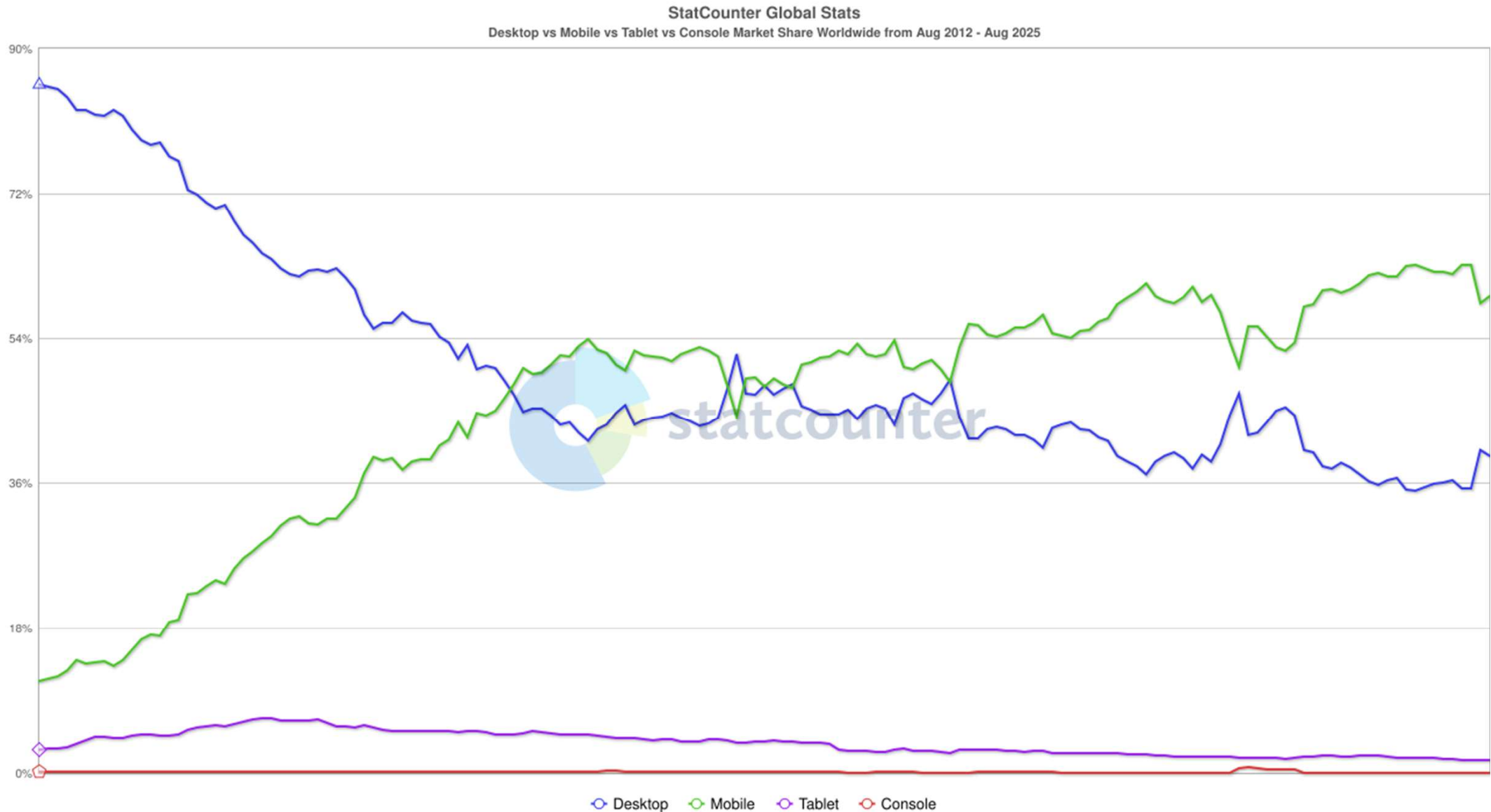
Mobile operating systems

- Operating system
 - Software layer that virtualizes the hardware, facilitating and maximizing its use
- Mobile operating system
 - It is an operating system suited to the specificities of mobile devices/platforms

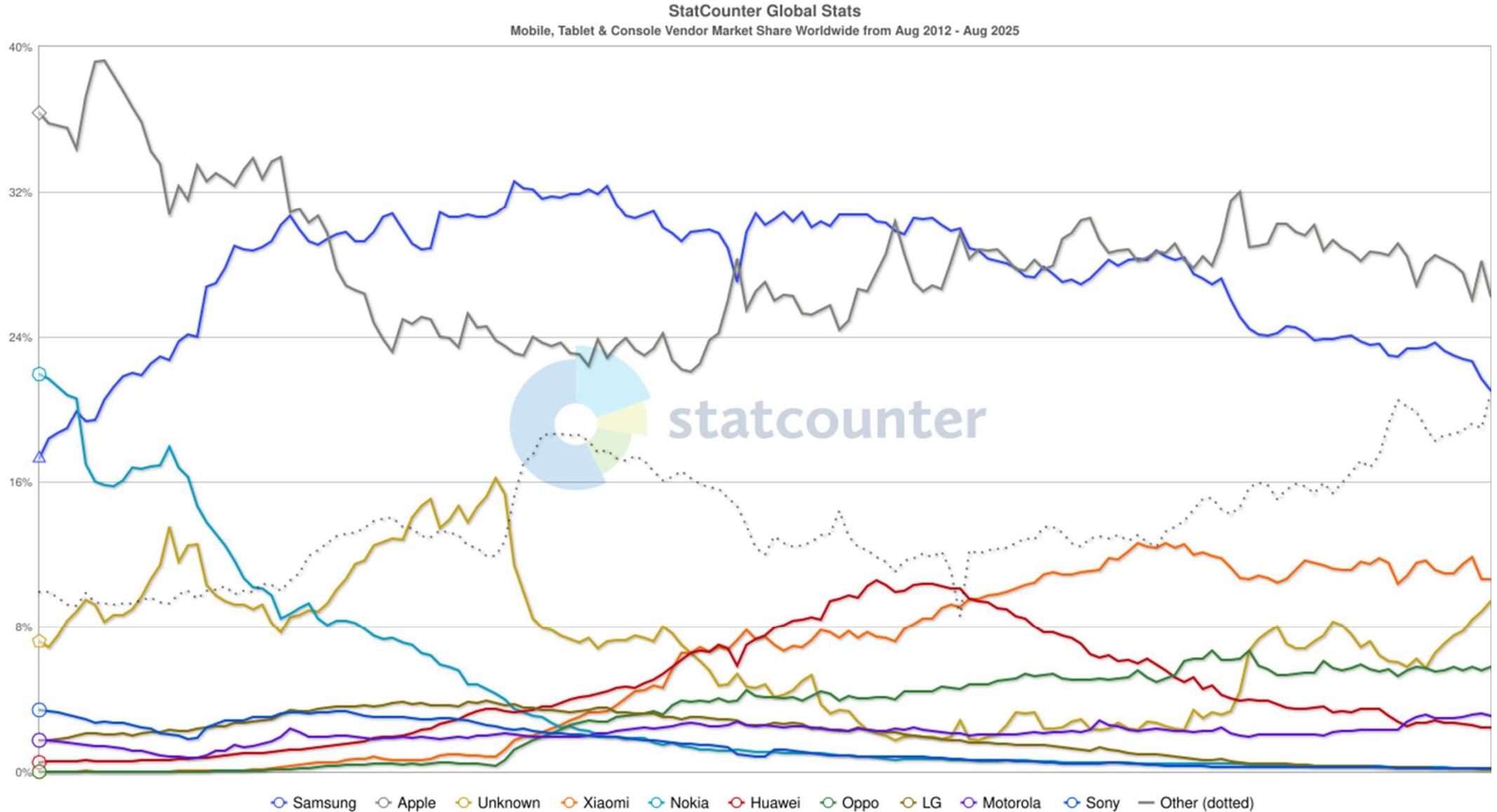


Market share

Mobile	Desktop	Tablet	Console
59.08%	39.35%	1.55%	0.03%



Manufacturers



Mobile operating systems

- Symbian OS
- RIM BlackBerry OS
- iOS (Apple)
- Windows CE
- Windows Mobile
- Windows Phone



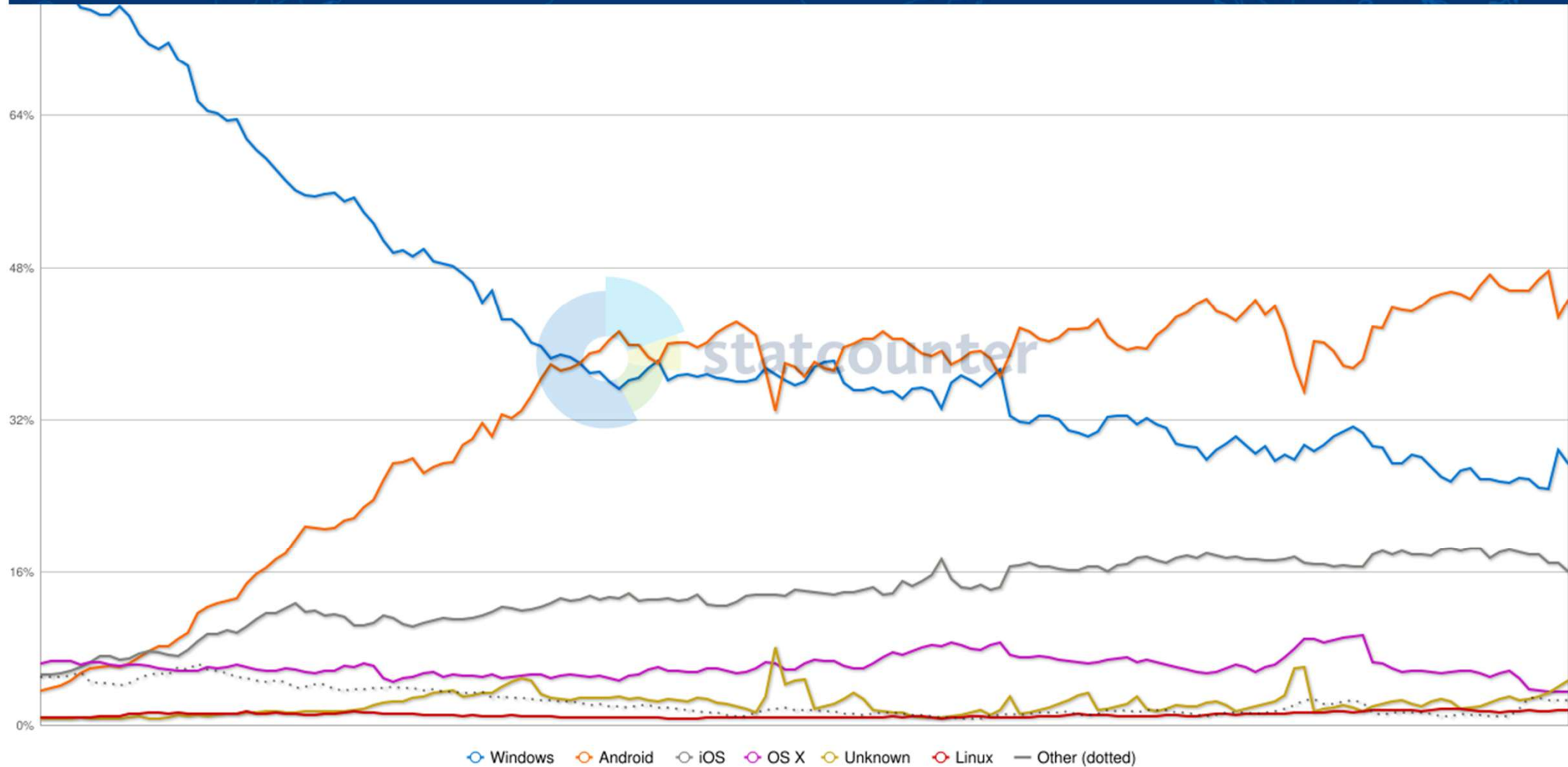
- Android
- Firefox OS
- KaiOS
- LiMo
- Maemo
- MeeGo
- Bada
- Mer
- Tizen
- Open Web OS/ webOS
- Jolla Sailfish OS/ Sailfish UI
- Ubuntu Touch
- ...
- HarmonyOS



Mobile OS

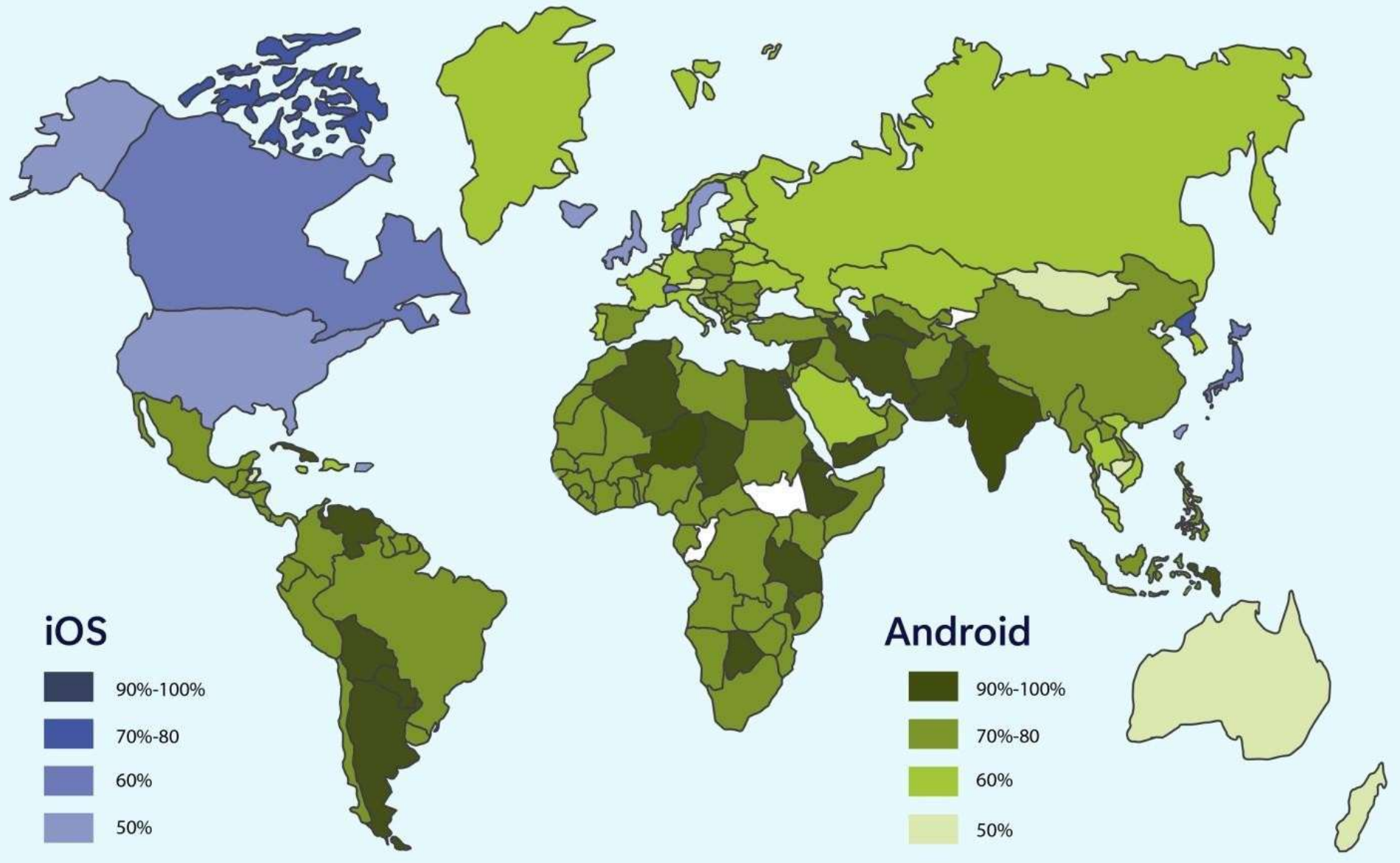
Feature	Android	iOS	HarmonyOS	Tizen	KaiOS	Sailfish OS Aurora OS	GrapheneOS	Ubuntu Touch	Mobian ^[1]	Plasma Mobile	PureOS	PostmarketOS
Developed by	Google, Open Handset Alliance	Apple Inc.	Huawei	Linux Foundation, Tizen Association, Samsung, Intel	KaiOS Technologies Inc.(TCL)	Sailfish Alliance, Mer, Jolla and Sailfish community contributors	GrapheneOS Foundation	UBports and Ubuntu community contributors (previously Canonical Ltd.)	Debian on Mobile Team	KDE and Blue Systems	Purism	PostmarketOS community
Market share ^[2]	77%	19%	4%	0.22%	0.14%	N/A	N/A	N/A	N/A	N/A	N/A	N/A
License	Partial; base system is open source , but many devices use proprietary drivers for hardware support, and most Android operating systems include Proprietary apps (such as Google Play and other Google apps). ^{[3][4]}	Proprietary , open source kernel and core	Proprietary except for open-source components	Partial; both proprietary and open-source components, assorted licenses	Proprietary except for open source kernel patches (formerly the MPL 2.0 B2G OS)	Partial; free and open-source , but the UI and the SDK are proprietary and closed source	Partial; free and open-source system with proprietary drivers	Free Free and open-source , mainly the GPL ^[5]	Free Free and open-source , mainly the GPL	Free Free and open-source , mainly the GPL	Free Free and open-source , mainly the GPL	Free Free and open-source , GPL
Current version	16	18.6	5.0.0.102(SP3C00E73R4P17) - Release	9.0 M2	3.1	4.5.0.24	2025070800	20.04 OTA-9	12.0	6.3.5	10.3	24.12
Development version	16 Beta 4	26 Beta 4	5.0.0.103 - Release (Beta)		3.2	Unknown	2025070800	Unknown	20250615	Unknown	Unknown	N/A
Current version release dates	September 3, 2024; 12 months ago	July 29, 2025; 37 days ago	October 23, 2024; 10 months ago	May 30, 2020; 5 years ago ^[6]	?	September 12, 2023; 23 months ago ^[7]	July 8, 2025; 58 days ago	June 3, 2025; 3 months ago	June 6, 2023; 2 years ago	May 6, 2025; 3 months ago	June 14, 2023; 2 years ago	December 23, 2024; 8 months ago

Worldwide



Dominant OS

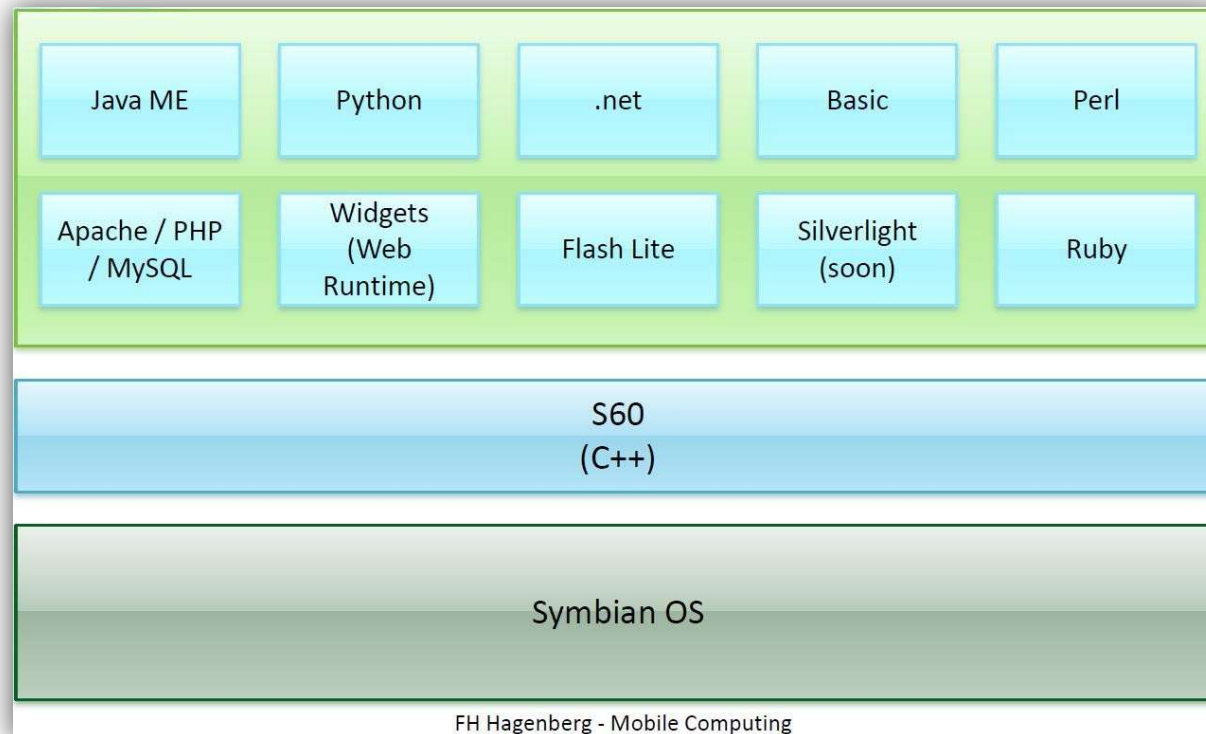
iOS vs Android market share 2023



- Evolution of PSION's EPOC system (1989-2000) designed for PDA
- *Symbian* was created in 1998
 - Joint venture between *Psion* and some mobile phone manufacturers (*Ericsson*, *Motorola* and *Nokia*)
- The first version of Symbian OS came out in 2001
 - Nokia 9210
- In 2008 Nokia acquired *Symbian Software Limited*
 - *Symbian Foundation* was created – autonomous and non-profit entity to manage Symbian OS
 - In November 2010, the project was abandoned, leaving only support and development for specific markets
- Preemptive, single-user, multitasking operating system with memory protection
- <http://www.symbian.org>

Symbian OS

- Development...
 - Programming was essentially carried out in C++
 - Symbian C++
 - Subsystems were developed to support other languages/development platforms



RIM BlackBerry

- RIM – *Research In motion*
 - Canada
 - 1996
- Main objective: “personal digital assistant”
 - Email
 - Calendar
 - Tasks
 - **Security**
- Development
 - Essentially JAVA
 - *Web Service extensions (Server + Client)*
- **Evolution (more recent years)**
 - Android system was adopted for their devices
 - Dedicated to business and security services

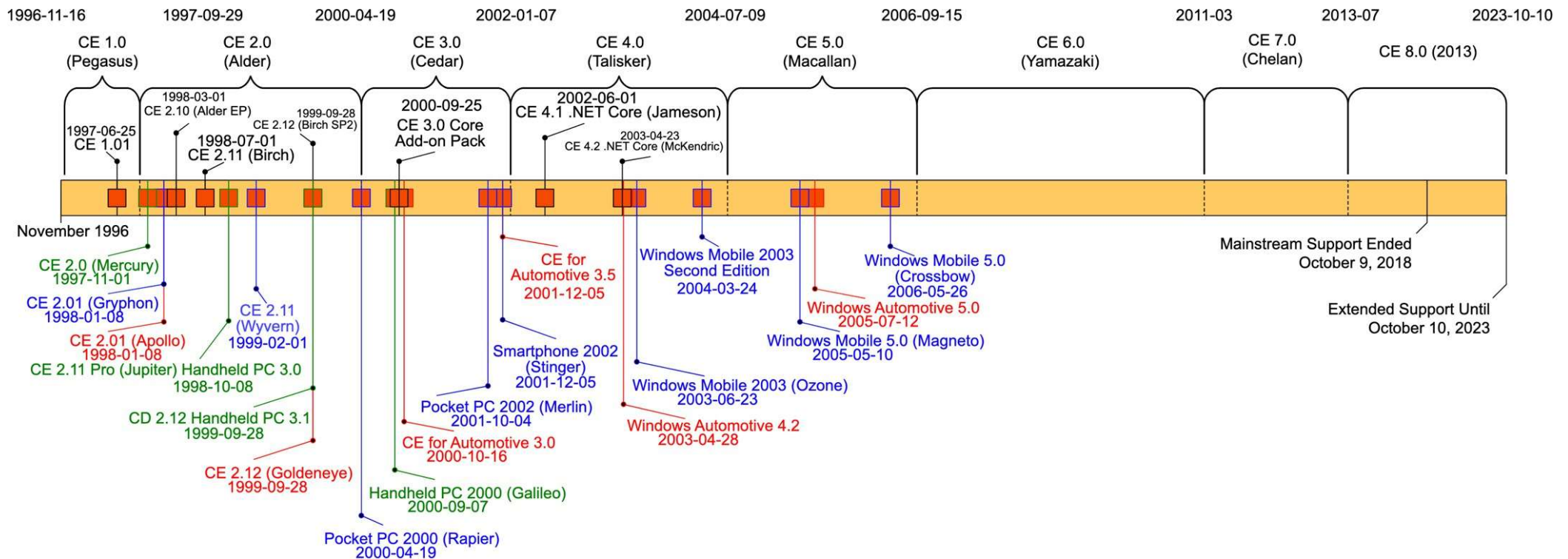


Windows CE



- Windows Embedded Compact
 - Some call it “Windows Compact Edition”
 - Microsoft says “Compact, Connectable, Compatible, Companion, and Efficient ”
- Developed by Microsoft for embedded systems and/or limited resources
 - Goal: Provide “similar” features to those found on a Windows computer
- 1st version: 1996
- Last version: 8
 - Windows Embedded Compact 2013

Windows CE



Windows Mobile

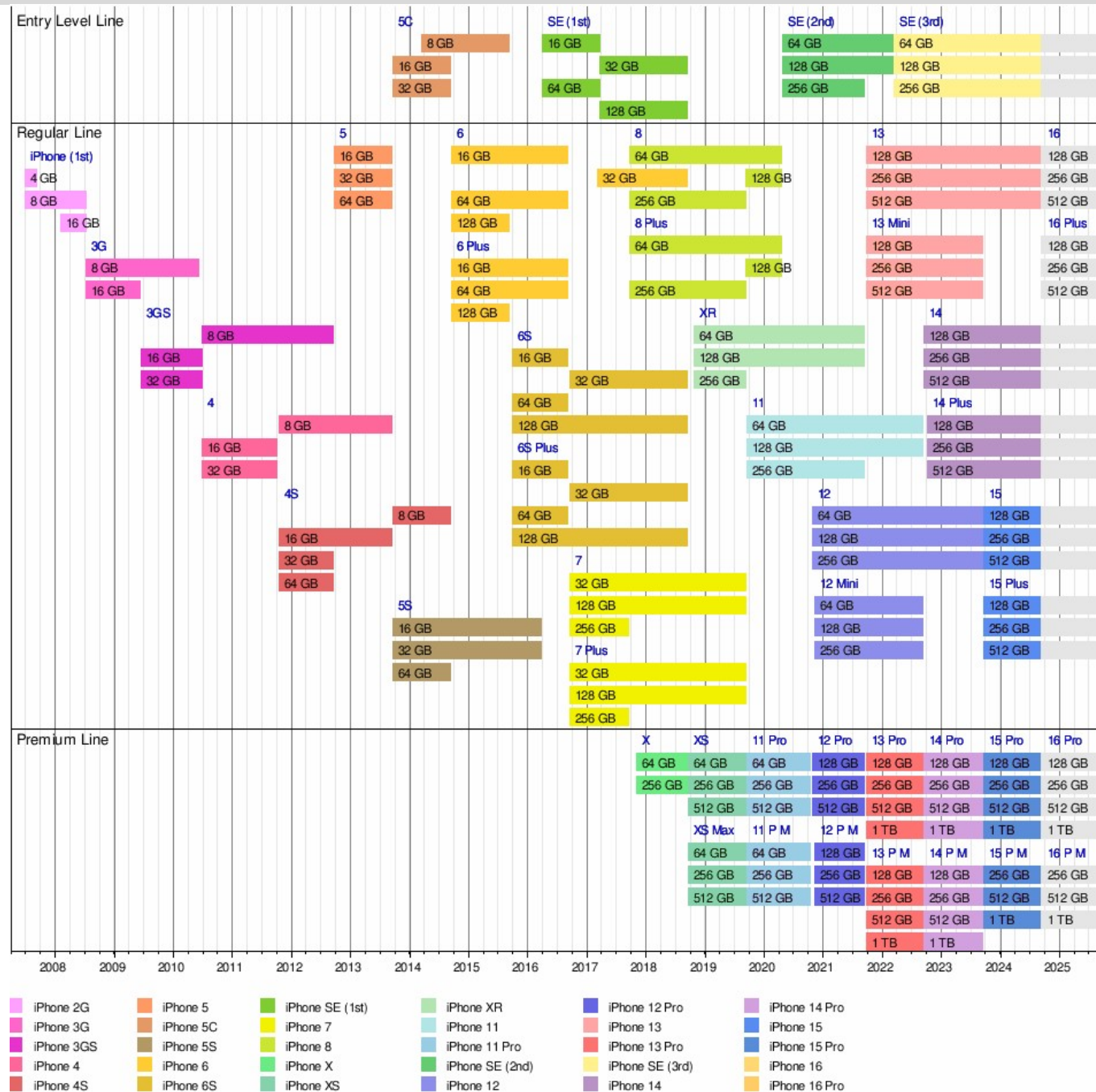
- Based on *Windows CE*
 - *Windows Mobile 6* is based on *Windows CE 5.2*
- Used on cell phones/*smartphones*
- *Standard* and *Pro* versions
 - The *Pro version* has touchscreen support
- Latest version: 6.5.5
- Development
 - Various languages
 - Initially, the most common was C/C++ (Win32-API, MFC)
 - After, development began to be supported through the .NET Compact Framework
 - Subset of the .NET Framework
 - Version of the .NET Framework adapted to devices with limited resources
 - Continues to offer services not normally associated with this type of equipment
 - For example, SQL – using the Microsoft SQL Server CE version
 - C#
 - VB.NET

Windows Phone

- *Windows Phone 7*
 - Based on *Windows CE 6.0*
- *Windows Phone 8 and 8.1*
 - Based on *Windows NT + Windows 8*
- *Windows 10 Mobile*
- Pleasant interface but failed to gain market share
- Main advantage: integration with business applications and Office
- Cortana assistant was introduced to respond to the competition (Siri, Google Now)
- Development
 - Based on two technologies
 - Silverlight
 - XNA
 - Development platform: Visual Studio
 - There was a version of Visual Studio Express prepared for developing applications for Windows Phone
 - Base language: C#
 - Applications were made available through the Windows Phone Marketplace/App hub
 - 30% to Microsoft
 - \$99/year

- Designed for *iPhone* and *iPod Touch* from Apple
 - Strong relationship with *hardware*
 - Simplicity of use
 - Stability
- iOS devices are considered the main reference in *smartphones* and *tablets*
- Initially based on OS X 10.5 (*Leopard*)
- Multitasking support began with the release of version 4
 - In previous versions it was already possible to multitask between some specific applications (e. g., music)
- Latest version released: iOS 18.6.2
 - *The iOS 26 and the new iPhones will be available soon*
 - <https://www.apple.com/apple-events/>
 - In recent versions, there has been an approximation in the provision of functionalities until now associated with other systems (mainly Android)
 - iPadOS appeared with the launch of iOS 13

iPhone



iOS (iPhone, iPad)

- Development
 - Applications distributed through the 'App Store'
 - Free applications
 - Paid applications
 - 30% to Apple
 - \$99/year to subscribe to iOS Developer Program
 - Necessary tools and requirements:
 - SDK
 - Xcode – development environment
 - Includes iPhone Simulator
 - Languages: C, C++, Objective-C, **Swift**, **SwiftUI**
 - Computer with macOS
 - iPhone, iPad, Apple Watch, Apple TV



Android is a trademark of Google LLC.

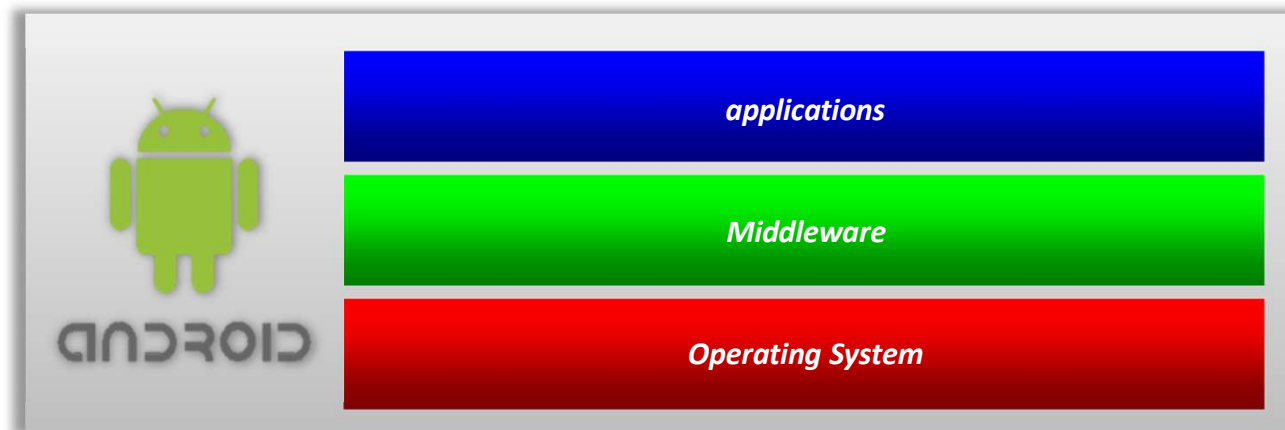
- Open-source initiative developed by a group of companies, headed by Google, which was named Open Handset Alliance

“Today, there are 1.5 billion television sets in use around the world. 1 billion people are on the Internet. But nearly 3 billion people have a mobile phone, making it one of the world's most successful consumer products” (November 5, 2007)

http://www.openhandsetalliance.com/oha_overview.html

Android

- Software suite that includes the operating system, *middleware* and applications
 - Linux-based (initially, version 2.6)
 - From version 4 of *Android*, version 3 of Linux started to be used
 - From version 7 of *Android*, version 4 of Linux started to be used
 - From version 11 of *Android*, version 5 of Linux started to be used
 - From version 14 of *Android*, version 6 of Linux started to be used
- Support for the latest applications and technologies in the area of mobile devices (photography, video, GPS, compass, accelerometer, games, OpenGL, SQLite, ... and cell phone 😊)



Android

- Google has followed a policy of launching its “own” device to support the presentation of a new version of the Android system (with significant changes)
 - *Nexus One* (manufactured by HTC)
 - The way chosen for commercialization undermined the success of the project
 - Android 2.1
 - *Nexus S* (made by Samsung)
 - The commercialization model followed in the second model was more successful
 - In addition to the innovations in terms of the Android system, it served to present a set of new technologies (e. g., NFC, curved screen)
 - Android 2.3
 - (Samsung) Galaxy Nexus
 - Android 4.0
 - Nexus 7 (Tablet - ASUS)
 - Android 4.1
 - Nexus 4 (LG)
 - Android 4.2
 - Nexus 7 2013 (Tablet - ASUS)
 - Android 4.3
 - Nexus 5 (LG)
 - Android 4.4



Android

- Nexus 6 (Motorola)
 - Android 5.0
- Nexus 9 (Tablet - LG)
 - Android 5.0
- Nexus 5X (LG)
 - Android 6.0
- Nexus 6P (Huawei)
 - Android 6.0



Android – Google Pixel phones

Model	Announcement	Release		Support ^[1]		Lifespan ^[a]
		Date	Launch OS	Date ^[2]	Final OS ^[2]	
Pixel / XL	October 4, 2016	October 20, 2016	7.1 Nougat	December 2, 2019	10	3 years, 1 month
Pixel 2 / XL	October 4, 2017	October 17, 2017	8.0 Oreo	December 7, 2020	11	3 years, 1 month
Pixel 3 / XL	October 9, 2018	October 18, 2018	9 Pie	June 6, 2022	12	3 years, 7 months
Pixel 3a / XL	May 7, 2019	May 15, 2019		September 5, 2022	12L	3 years, 3 months
Pixel 4 / XL	October 15, 2019	October 23, 2019	10	February 6, 2023	13	3 years, 3 months
Pixel 4a	August 3, 2020	August 20, 2020		November 6, 2023		3 years, 2 months
Pixel 4a (5G)		November 5, 2020	11	February 5, 2024	14	3 years, 3 months
Pixel 5	September 30, 2020	October 15, 2020		November 4, 2024		3 years, 3 months
Pixel 5a	August 17, 2021	August 26, 2021				3 years, 2 months
Pixel 6 / Pro	October 19, 2021	October 28, 2021	12	At least October 2026	—	3 years, 10 months
Pixel 6a	May 11, 2022	July 21, 2022		At least July 2027		3 years, 1 month
Pixel 7 / Pro	October 6, 2022	October 13, 2022	13	At least October 2027		2 years, 10 months
Pixel 7a	May 10, 2023	May 10, 2023		At least May 2028		2 years, 3 months
Pixel Fold		June 20, 2023		At least June 2028		2 years, 2 months
Pixel 8 / Pro	October 4, 2023	October 12, 2023	14	At least October 2030		1 year, 10 months
Pixel 8a	May 7, 2024	May 14, 2024		At least May 2031		1 year, 3 months
Pixel 9 / Pro / Pro XL	August 13, 2024	August 22, 2024		At least August 2031		1 year
Pixel 9 Pro Fold		September 4, 2024		At least September 2031		1 year
Pixel 9a	March 19, 2025	April 10, 2025	15	At least March 2032		4 months
Pixel 10	August 20, 2025	August 28, 2025	16	At least August 2032		0 months
Pixel 10 Pro / Pro XL		August 28, 2025				0 months
Pixel 10 Pro Fold		October 9, 2025		At least October 2032		

Versions

List of Android Versions and Initial Stable Release Dates				
 Android 1.0 September 23, 2008	 1.5 - Cupcake April 27, 2009	 1.6 - Donut September 15, 2009	 2.0/2.1 - Éclair October 26, 2009	
 2.2 - Froyo May 20, 2010	 2.3 - Gingerbread December 6, 2010	 3.0 - Honeycomb February 22, 2011	 4.0 - Ice Cream Sandwich October 18, 2011	
 4.1/4.3 - Jelly Bean July 9, 2012	 4.4 - KitKat October 31, 2013	 5.0 - Lollipop November 12, 2014	 6.0 - Marshmallow October 5, 2015	 7.0 - Nougat August 22, 2016
 8.0 - Oreo August 21, 2017	 9.0 - Pie August 6, 2018	 Android 10 September 3, 2019	 Android 11 September 8, 2020	 Android 12 October 17, 2021





2022.08.15



2023.10.04



2024.09.03

Versions

ANDROID PLATFORM VERSION		API LEVEL	CUMULATIVE DISTRIBUTION
4.4	KitKat	19	
5	Lollipop	21	99.9%
5.1	Lollipop	22	99.8%
6	Marshmallow	23	99.3%
7	Nougat	24	98.6%
7.1	Nougat	25	98.0%
8	Oreo	26	97.4%
8.1	Oreo	27	96.4%
9	Pie	28	93.4%
			87.6%
10	Q	29	
			77.4%
11	R	30	
			61.5%
12	S	31	
			48.7%
13	T	33	
			31.9%
14	U	34	
15	V	35	4.5%

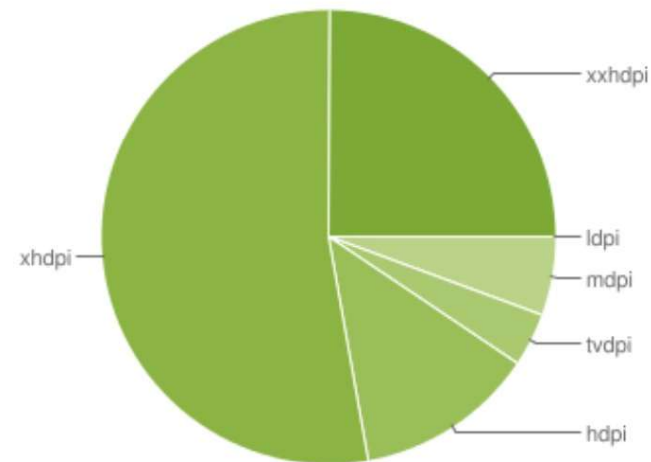
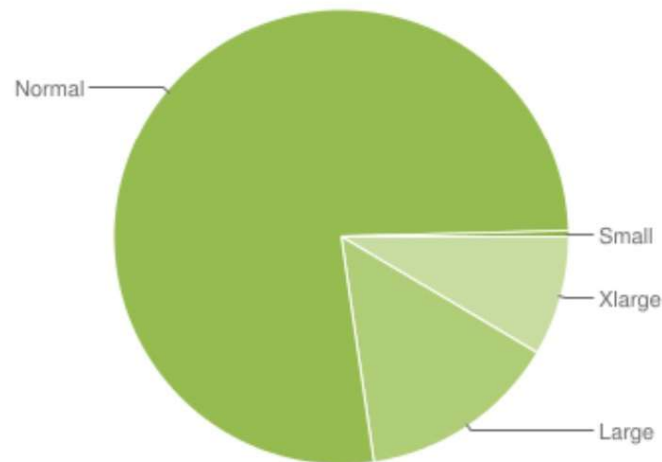
Last updated: April 1, 2025

• Android versions for different devices:

- Android OS
 - Smartphones, tablets and similar devices
- WearOS
 - Smartwatches and similar
- Android TV
 - TV, streaming devices, set-top boxes and large screens
- Android Automotive/Auto
 - AndroidCar
- ~~Android Things~~
 - ~~IoT~~

Screen resolution

	ldpi	mdpi	tvdpi	hdpi	xhdpi	xxhdpi	Total
Small					0.4%		0.4%
Normal		0.1%	0.3%	7.9%	45.4%	23.2%	76.9%
Large		1.2%	3.4%	1.1%	6.8%	1.7%	14.2%
Xlarge		4.3%	0.1%	3.8%	0.3%		8.5%
Total	0.0%	5.6%	3.8%	12.8%	52.9%	24.9%	



Data collected during a 7-day period ending on January 6, 2023.

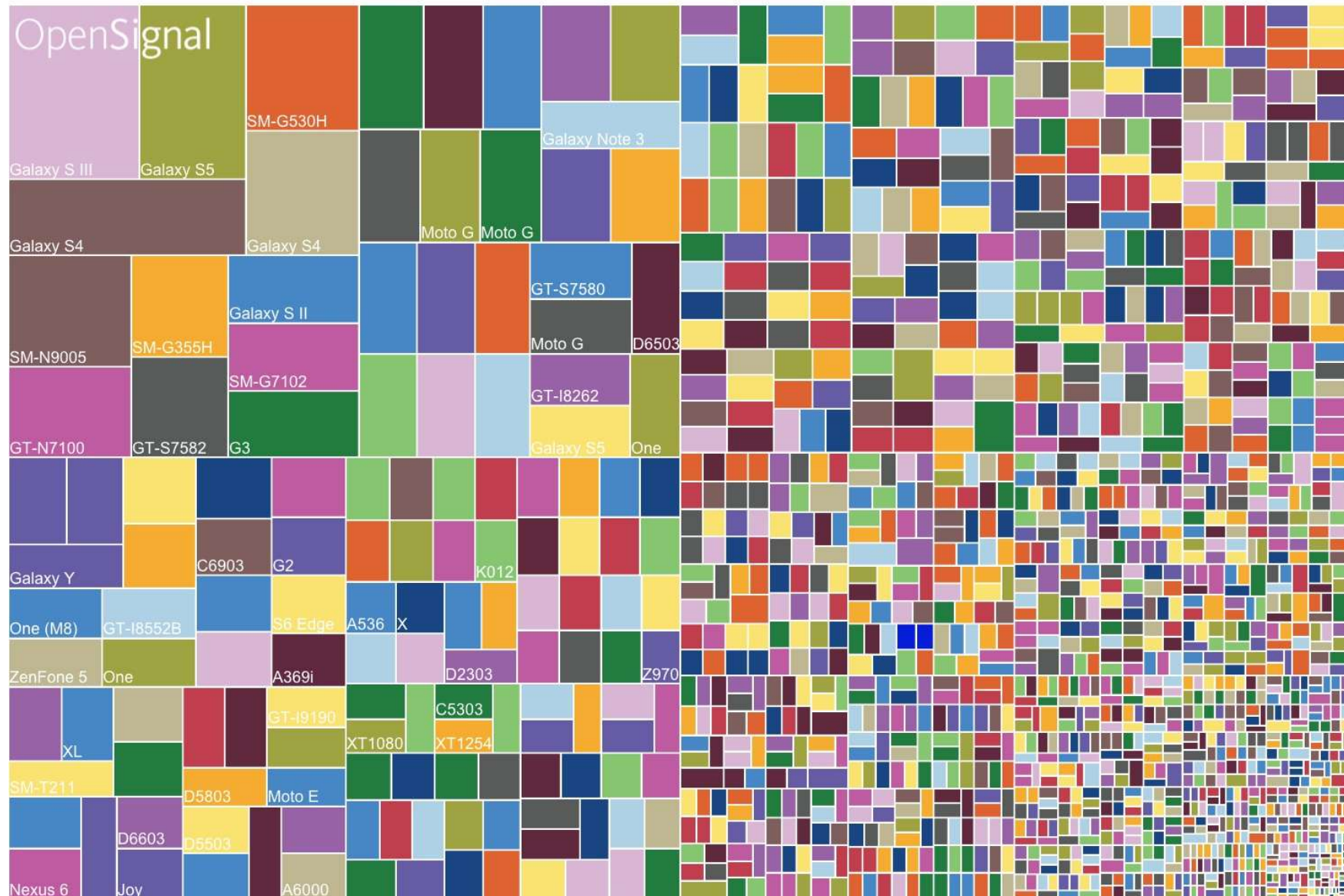
Any screen configurations with less than 0.1% distribution are not shown.

Android Fragmentation

- The problem of fragmentation, usually associated with the Android system, which makes it difficult to create new applications compatible with the various existing devices, arises due to several reasons:
 - Different versions of Android active at the same time
 - Different device manufacturers, with different operating system update policies
 - Economic context of the Android customer opting for cheaper devices
 - Different hardware
 - Screen sizes and resolutions
 - Processors
 - Memory capacity
 - Sensors
 - Communication technologies

Android Fragmentation

DEVICE FRAGMENTATION

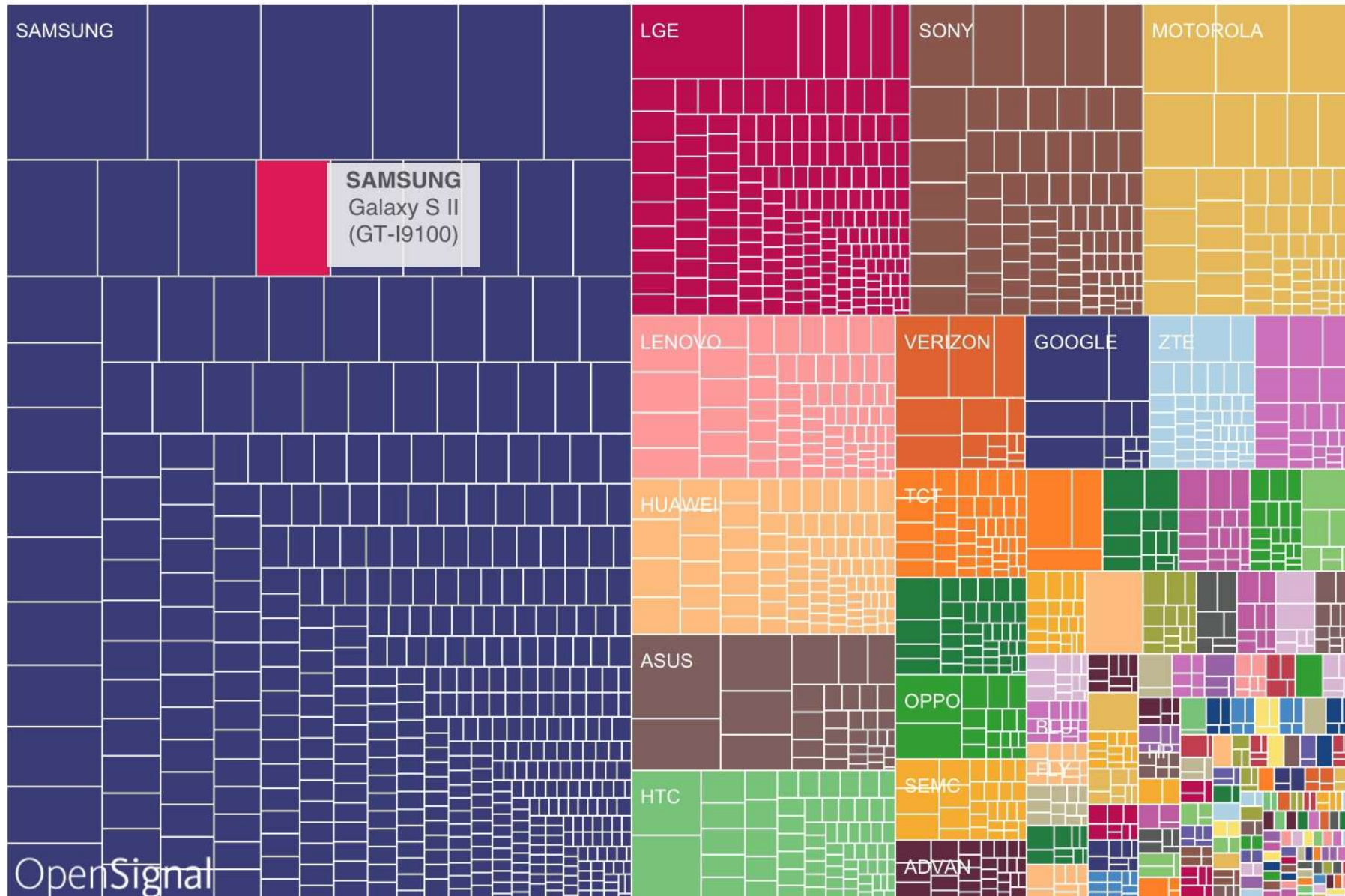


August 2014

August 2015

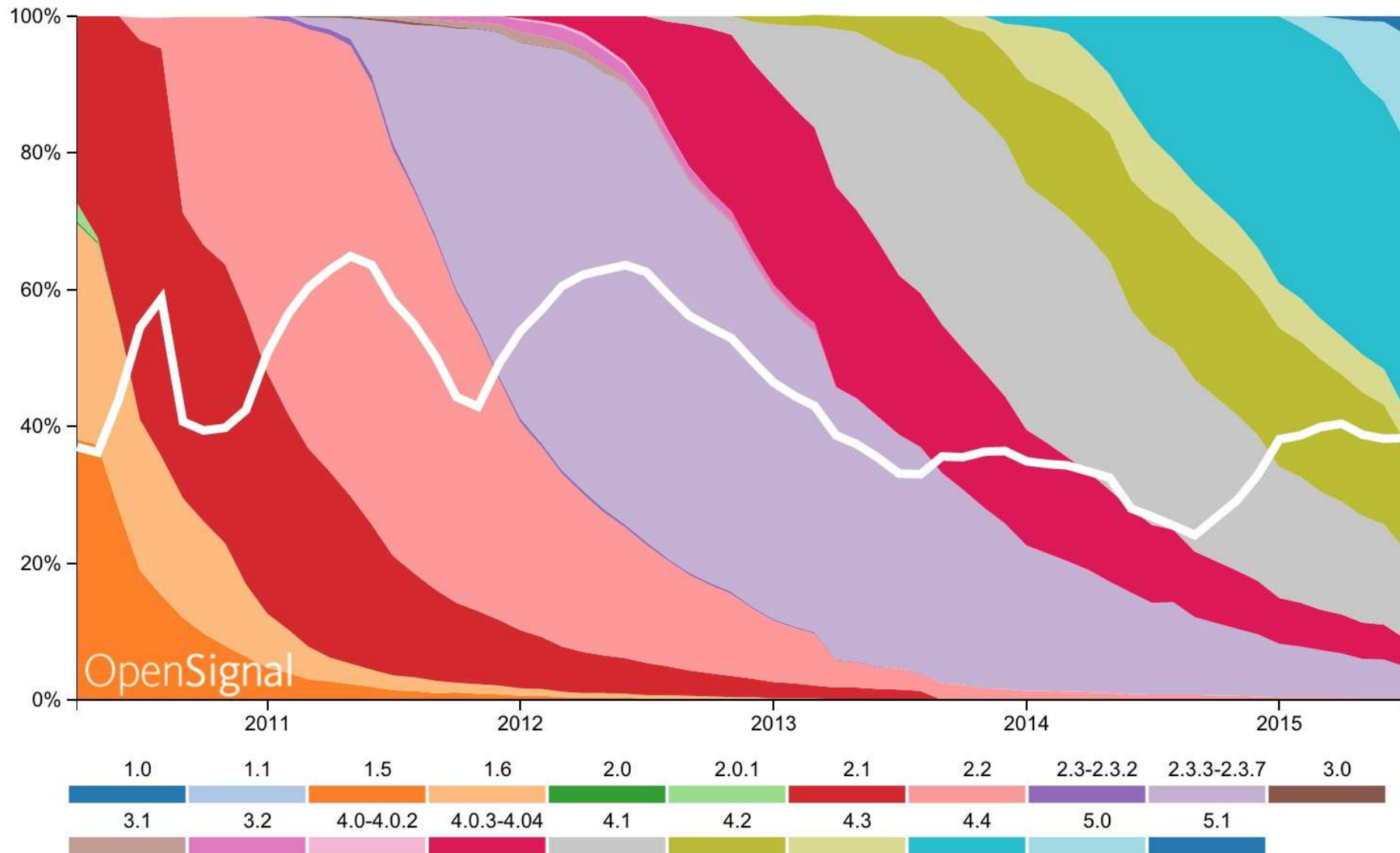
Android Fragmentation

BRAND FRAGMENTATION



Android Fragmentation

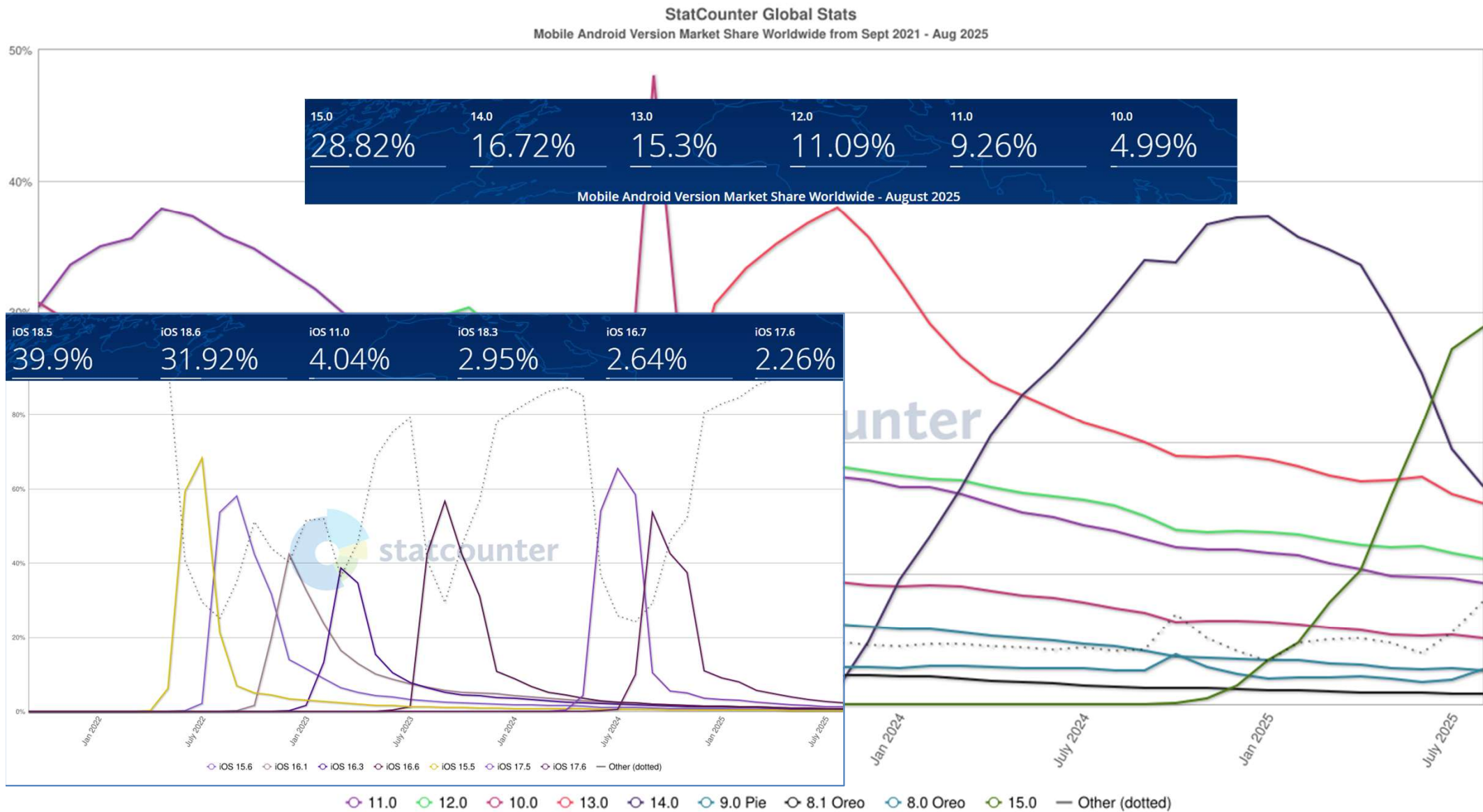
ANDROID OPERATING SYSTEM FRAGMENTATION



The white line shows the market share of the leading API level at any time

Android Fragmentation

<https://gs.statcounter.com/android-version-market-share/mobile/worldwide/#monthly-202109-202508>



Fragmentation

- Considerations for application development:
 - Different screen dimensions
 - Opt for creating dynamic screens rather than using fixed dimensions
 - Different memory capacities
 - Consider using lower resolution images to support older devices or providing multiple versions of images (with varying resolutions) within the application
 - Different processing capacities
 - Avoid heavy processing within the application; redirect such tasks to online processing whenever possible
 - Dependence on sensors and communication technologies
 - Accelerometers, Gyroscopes, etc.
 - Mobile data, Wi-Fi, Bluetooth, Bluetooth BLE, NFC, etc.
 - Diversity of target audience
 - Account for language and interfaces compatible with RTL (Right-To-Left) text, among other considerations

Android architecture



Android development

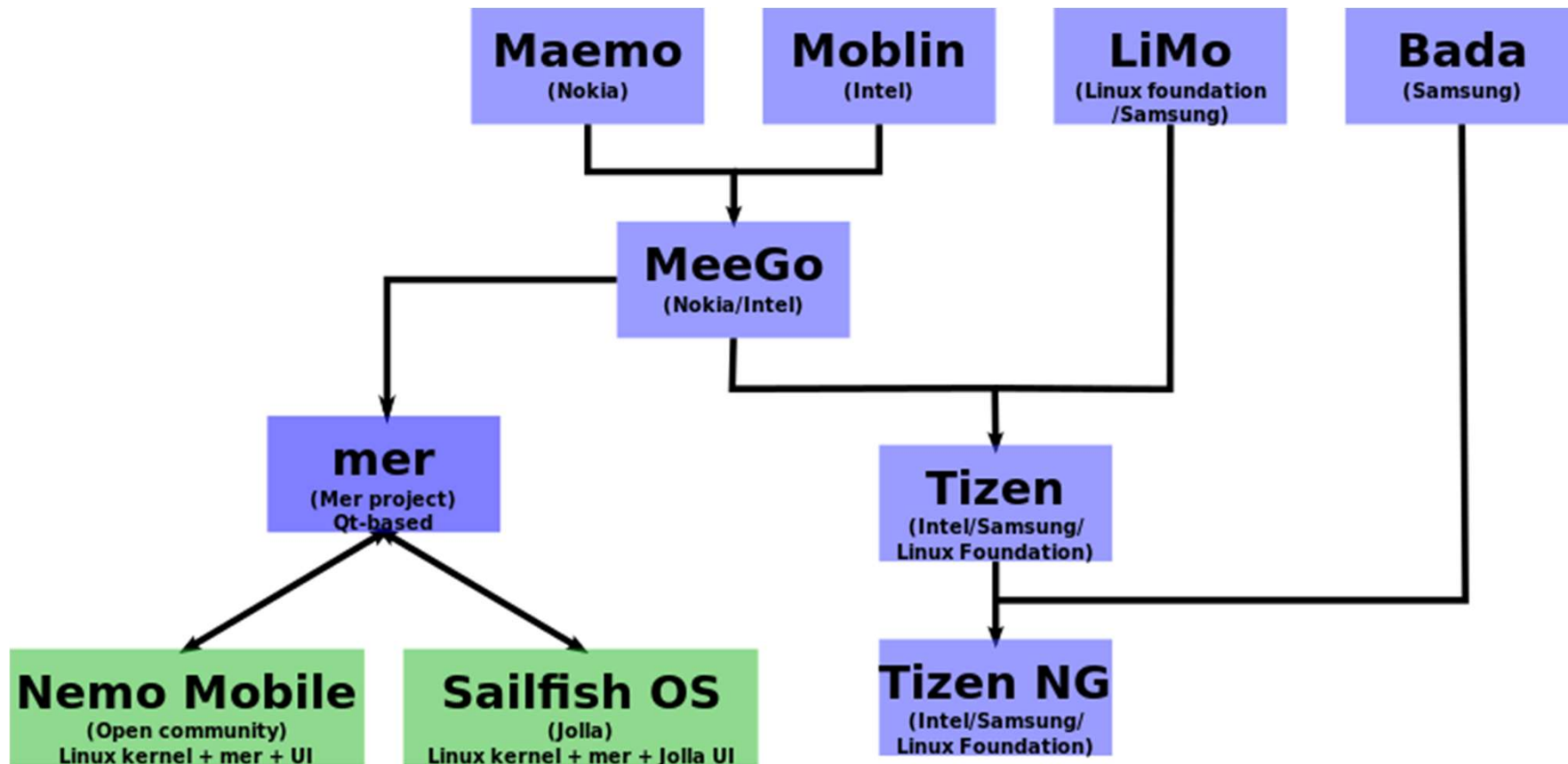
- Java, **Kotlin** and C/C++
 - A SDK is provided (Android SDK)
 - IDE: Android Studio, based on IntelliJ by JetBrains
 - Includes emulators for the various versions of Android
- Application distribution
 - packages ".apk"
 - For more recent versions it is mandatory to use bundles ".aab"
 - Google Play Store and other “parallel” markets
 - Packages copied directly to the cell phone can be installed
 - Google Play Store registration has a \$25 one-time fee
 - Fee of 30% of the price of each application sold

- LiMo – *Linux Mobile*
- Management and development by the *LiMo Foundation*
- Essentially focused on *middleware*
- The platform is not *open source* but it is all based on *open source modules*
- Programming/development technologies
 - Java
 - HTML + Javascript
 - C/C++
 - Flash



Tizen , Sailfish ,...

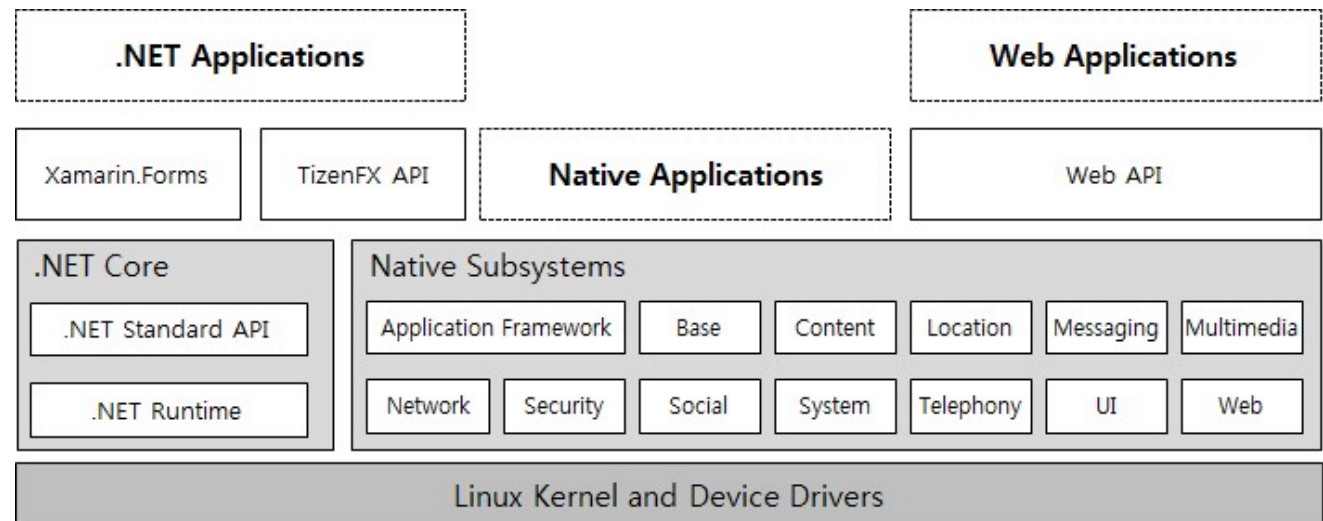
- LiMo + Meego + ... => Tizen



- Jolla devices
- Development carried out in:
 - QML (Qt Framework)
 - C++



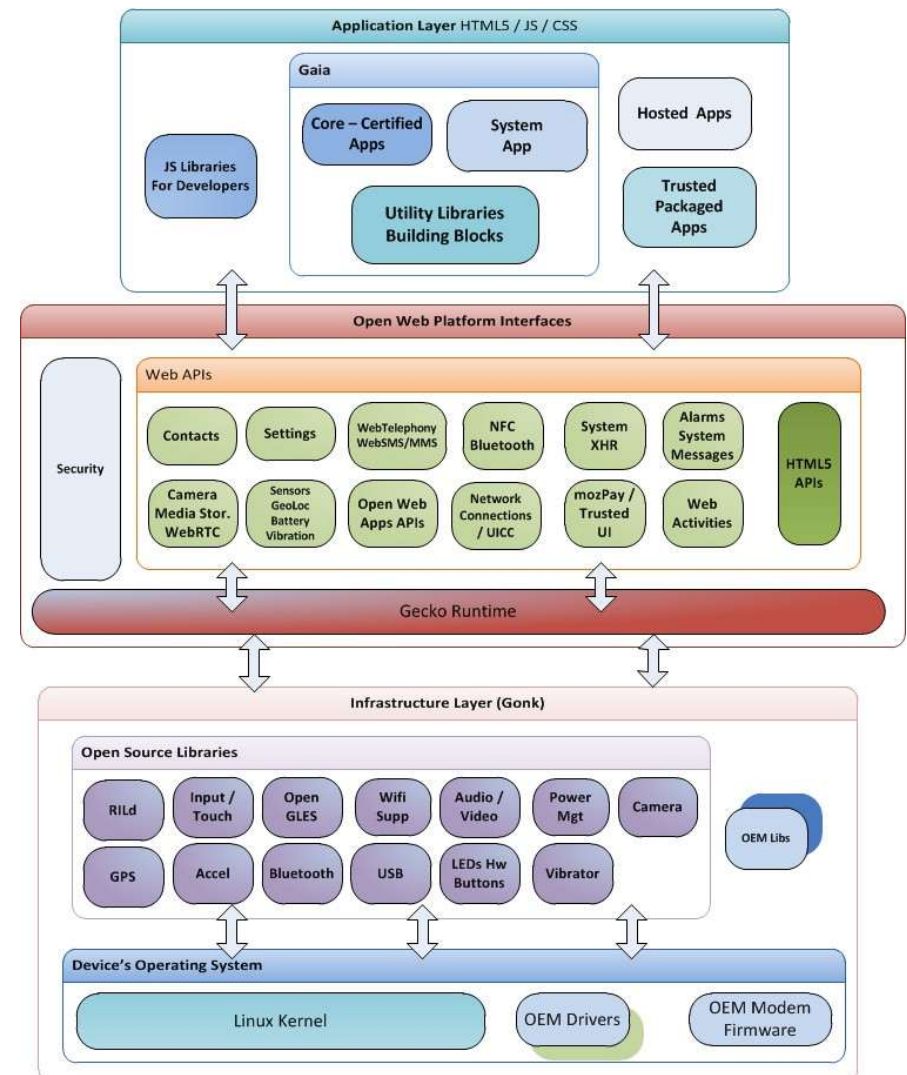
- Project supported by the Linux Foundation
- Development based on
 - Linux kernel
 - GNU C libraries
- Applications developed using
 - Web (HTML5, CSS, JavaScript)
 - Native
 - .NET Applications (C#)



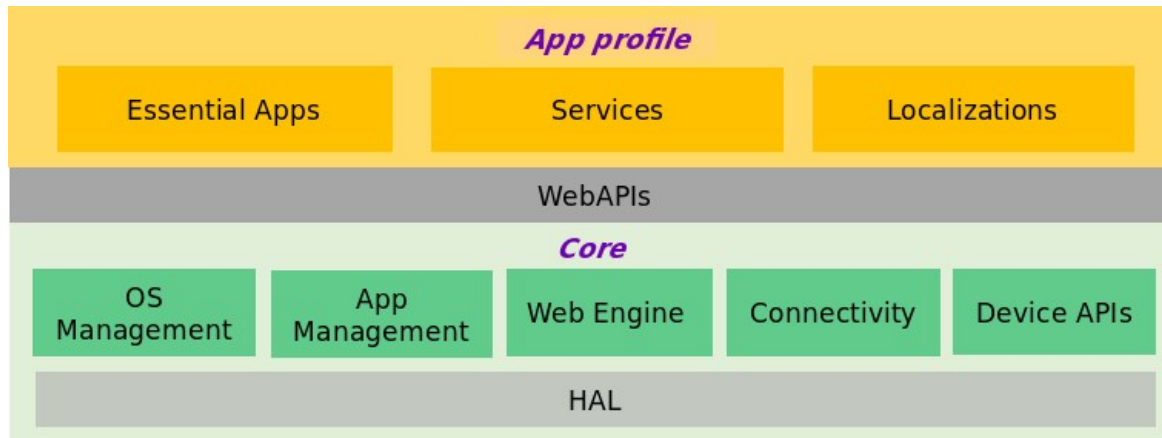
FirefoxOS



- Mainly web-oriented system
- Use of existing Web standards to provide applications
 - HTML5
 - JavaScript
- Use WebAPIs to access resources available on the hardware
- 1st version: August 2012
- Latest version: May 2016 (version 2.6)



- Linux-based
- Designed for lower-cost cell phones, normally with associated keyboards
- 1st version: March 2017
- Latest version: 4 (May 2025)



Cross-platform applications

- Main goal
 - Develop only one version of the application that will run on different operating systems
 - Faster development



Cross-platform

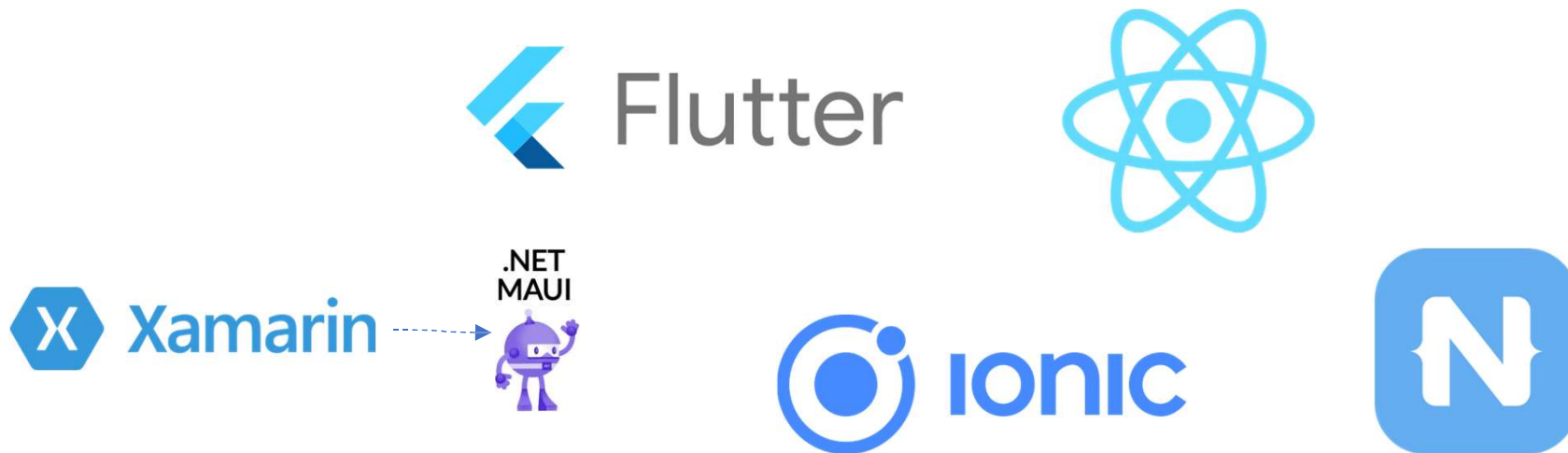
- Benefits
 - Same development language for all systems
 - It is not necessary to have specialized development teams for each system
 - Teams focused on providing new features rather than adapting to different systems
 - Application support made easy
 - Larger target audience

Cross-platform

- Disadvantages
 - Possibly lower application performance
 - Limited access (possibly) to internal equipment resources and sensors
 - Difficulty adapting interfaces to a greater diversity of screen sizes and interaction systems
 - Related to the fragmentation problem mentioned earlier, although existing platforms try to minimize them

Cross-platform

- Until recently, the main solution was the Java language
- There are currently several cross-platform solutions on the market
 - Cross-platform development environments
 - With native code generation for different platforms
 - With object code generation that runs in the context of a pre-installed framework on devices
 - Based on the capabilities of Javascript and HTML5
 - Web Applications
 - Hybrids (native + web)



Cross-platform

	 React Native	 ionic	 Xamarin	 Flutter
	A framework for building native apps with React.	One app running on everything.	Complete Binding for the underlying SDKs.	Makes it easy and fast to build beautiful mobile apps.
	77 401 	77 401 		64 773 
Developers	Facebook	Drifty	Microsoft	Google
Language	JavaScript	TypeScript	C#	Dart
Performance	Close to native	Moderate	Moderate	Amazingcontrolle
GUI	Use native UI controllers	HTML, CSS	Use native UI controllers	Use proprietary widgets and deliver UI
Code reusability	90% of code is reusable	98% of code is reusable	98% of code is reusable	50 - 90% of code is reusable
Testing	Mobile device or emulator	Any browser	Mobile device or emulator, test cloud	Mobile device or emulator
Hot reload	Yes	No	No	Yes
Apps	Airbnb, Discord, Instagram	MarketWatch, Pacifica, JustWatch	Olo, Storyo, APX	KlasterMe, PostMuse Reflectly